Milestone M3 (Lane B): Comprehensive Code, Mathematical & Workspace Audit Report

Project: IIT/SRIC/R/AEH/2026/104 (CMPDI) — *Dynamic Modelling and Digital Twin of Electric Cable Shovel* **Audited Codebase:** m3_observer_suite.m (v2.0.0-PRELIMINARY), run_m3_experiments_v2.py (v2.0.0-PRELIMINARY) **Validation Status: PRELIMINARY** (Verified against frozen M2 Simscape .mat archives P01–P04b and P08; awaiting Simulink MS02P05 pose-sweep .slx runs before promotion to FINAL)

1. Workspace Organization & Version Preservation (M3/00_README – M3/10_ARCHIVE)

All discovered M3 files (v1.0 synthetic baseline and v2.0.0-PRELIMINARY archive-verified suite) have been organized into the canonical 11-folder M3 hierarchy (M3_FILE_MIGRATION_REGISTER.xlsx) with **zero file deletions**:

- M3/00_README/M3_README.md: Architecture overview and directory map.
- M3/01_THEORY/M3_Theory_and_Derivations.md: Complete Euler--Lagrange momentum identity (\dot{D} = C + C^T), continuous--discrete RK4/FOH discretization, 3-tier Branch B uncertainty bound, and planar Jacobian J_s(q) = R(q_3)\Pi\,\mathrm{diag}(s(d_4), 1) derivations.
- M3/02_BRANCH_A_OBSERVER/m3_observer_suite.m: Friction-aware continuous--discrete RK4 (M=67) momentum observer.
- M3/03_BRANCH_B_OBSERVER/m3_observer_suite.m: Friction-uncertainty exact FOH momentum observer with 3-tier bound verification.
- M3/04_COMMON_CORE/run_m3_experiments_v2.py: Pure-Python MATLAB v7.3 HDF5 binary reader (MatV73Reader) and deterministic Park--Miller RNG (seed = 202603).
- M3/05_EXPERIMENTS/M3_VALIDATION_PROTOCOL.md: Complete 8-test validation protocol (A, B1, B2, C, D, E, F, G) plus 20-archive held-out cross-trajectory validation (P08).
- M3/06_RESULTS_MAT/: M3_observer_audit_results.mat and m3_audit_results.json.
- M3/07_FIGURES/: fig1_architecture_reversal.png through fig5_jacobian_conditioning.png.
- M3/08_REPORTS/M3_LaneB_Momentum_Observer_Report.pdf: Audited 9-page PDF report.
- M3/09_AUDIT/: M3_Code_to_Report_Discrepancy_Log.md and M3_CODE_AUDIT_REPORT.md.
- M3/10_ARCHIVE/v1.0_synthetic/: Archived v1.0 scripts, JSON, and HTML report preserved for full audit traceability.

2. Mathematical Verification of Euler--Lagrange

Skew-Symmetry (\dot{D} = C + C^T)

From shovel_params.m and equations_master.tex, the inertia matrix D(q) and Christoffel Coriolis matrix C(q,\dot{q}) with r(d_4) = d_4 + c = d_4 - 1.32\ \mathrm{m} are:
D(q) = \begin{bmatrix} I_{zz} + M_d r^2 & 0 \\ 0 & M_d \end{bmatrix}, \qquad C(q,\dot{q}) = \begin{bmatrix} M_d r \dot{d}_4 & M_d r \dot{q}_3 \\ -M_d r \dot{q}_3 & 0 \end{bmatrix}.
Differentiating D(q) with respect to time gives:
\dot{D}(q,\dot{q}) = \begin{bmatrix} 2 M_d r \dot{d}_4 & 0 \\ 0 & 0 \end{bmatrix}.
Summing C(q,\dot{q}) and its transpose C(q,\dot{q})^T:
C(q,\dot{q}) + C(q,\dot{q})^T = \begin{bmatrix} M_d r \dot{d}_4 & M_d r \dot{q}_3 \\ -M_d r \dot{q}_3 & 0 \end{bmatrix} + \begin{bmatrix} M_d r \dot{d}_4 & -M_d r \dot{q}_3 \\ M_d r \dot{q}_3 & 0 \end{bmatrix} = \begin{bmatrix} 2 M_d r \dot{d}_4 & 0 \\ 0 & 0 \end{bmatrix} \equiv \dot{D}(q,\dot{q}).
Consequently, \dot{D} - 2C = C^T - C = \begin{bmatrix} 0 & -2M_d r\dot{q}_3 \\ 2M_d r\dot{q}_3 & 0 \end{bmatrix} is strictly skew-symmetric, and the generalized momentum p = D(q)\dot{q} satisfies:
\dot{p} = \dot{D}\dot{q} + D\ddot{q} = (C + C^T)\dot{q} + (\tau_a + \tau_L - C\dot{q} - G(q) - F_v\dot{q} - F_c(\dot{q})) = \tau_a + \tau_L - G(q) - F_v\dot{q} - F_c(\dot{q}) + C(q,\dot{q})^T\dot{q},
where C(q,\dot{q})^T\dot{q} = \begin{bmatrix} 0 \\ M_d r \dot{q}_3^2 \end{bmatrix} (zero on the rotational saddle channel q_3, and positive centrifugal force +M_d r\dot{q}_3^2 on the prismatic crowd channel d_4).

Milestone M3 (Lane B): Code-to-Report Audit & Discrepancy Log

Project: IIT/SRIC/R/AEH/2026/104 — 2-DOF Electric Cable Shovel Dipper–Handle Digital Twin
Audit Version: v1.0.0 \rightarrow v2.0.0-PRELIMINARY
Validation Status: PRELIMINARY (Pending frozen MS02P05 / 05_configuration_sweep Simulink execution)

1. Executive Summary of Audit Findings

A line-by-line code-to-report audit was conducted across the initial Milestone M3 MATLAB script (m3_observer_suite.m v1.0), the numerical experiment runner (run_m3_experiments.py v1.0), the initial technical report (M3_LaneB_Momentum_Observer_Report.pdf v1.0), and the primary repository files in shovel-dynamics_m2 (shovel_params.m, recover_load.m, M2_null.mat, M2_100kN_0p1111Hz_sp1p50.mat, and stage1_results.mat).
All seven audit issues identified in the review have been resolved in **m3_observer_suite.m (v2.0.0-PRELIMINARY)** and **M3_LaneB_Momentum_Observer_Report.pdf (v2.0.0-PRELIMINARY)**:

Audit ID	Audit Category	Root Cause in v1.0	Resolution in v2.0.0-PRELIMINARY
AUD-01	**Data Ingestion & Validation**	build_reference_trajectory(mat_filepath, P) in	Replaced with load_and_validate_mat

Audit ID	Audit Category	Root Cause in v1.0	Resolution in v2.0.0-PRELIMINARY
		m3_observer_suite.m v1.0 accepted mat_filepath (defaulting to 'stage1_results.mat'), never called load(mat_filepath), and reconstructed a 28 s piecewise step trajectory (d_4 \in [2.6, 4.1]\ \mathrm{m}). Meanwhile, stage1_results.mat (1{,}020\ \mathrm{bytes}) is an M1 summary struct (results.ident, results.params_used) containing zero time-series vectors, while run_m3_experiments.py v1.0 fell back to an analytical sinusoid when scipy.io.loadmat failed on MATLAB v7.3 HDF5 files (M2_null.mat, M2_100kN_0p1111Hz_sp1p50.mat).	_archive(filepath, P) in m3_observer_suite.m and a native HDF5 v1 binary reader (MatV73Reader) in run_m3_experiments_v2.py that directly ingests M2_null.mat (SHA-256: 74a2ed3f...) and M2_100kN_0p1111Hz_sp1p50.mat (SHA-256: 6050507d...). Validates all 20 R.sig arrays, units, monotonicity, uniform 100\ \mathrm{Hz} sampling, and parameter lock against shovel_params.m. Passing stage1_results.mat or any missing/incomplete file now fails immediately with m3_observer_suite:SummaryOnlyMatFile / FileNotFound. Synthetic trajectory generation is isolated to opt-in 'Mode', 'synthetic_unit_test'.
AUD-02	**Sign Convention & Load Channel Mapping**	Report v1.0 §1.2 stated \tau_L = -Q_{\mathrm{ext}} = e_\tau = \text{ref}, whereas comparing D\ddot{q} + C\dot{q} + G + F_v\dot{q} + F_c = \tau_a + \tau_L against recover_load.m line 34 (D\ddot{q} + C\dot{q} + G + f = \tau_{\mathrm{actuator}} + Q_{\mathrm{ext}})	Corrected sign reconciliation in §1.2 and code so \tau_L \equiv +Q_{\mathrm{ext}} = [\text{R.sig.Qq3_pred}; \ \text{R.sig.Qd4_pred}] = -e_\tau = -[\text{R.sig.ref3};\ \text{R.sig.ref4}]. Also added **Check 0 (run_raw_slx_diagnos**

Audit ID	Audit Category	Root Cause in v1.0	Resolution in v2.0.0-PRELIMINARY
		requires \tau_L = +Q_{\mathrm{ext}} = +J_s(q)^T F = [\text{R.sig.Qq3_pred}; \ \text{R.sig.Qd4_pred}] = -e_\tau = -\text{ref}. Furthermore, v1.0 used synthetic uncoupled sinusoids (245\ \mathrm{kN\cdot m}, 49\ \mathrm{kN}) instead of the actual Jacobian-coupled loads in M2_100kN_0p1111Hz_sp1p50.mat (peak 734.24\ \mathrm{kN\cdot m} and 86.52\ \mathrm{kN}).	**tic)** documenting that P.model_has_coulomb = false in shovel_params.m line 50 during the 2026-08-25 Simulink runs.
AUD-03	**Report/Code Consistency (B1 vs. B2)**	Report v1.0 tables listed both B1: Constant (Reversing) and B2: Constant (Unidirectional), but m3_observer_suite.m v1.0 only implemented a single 'B_Constant' entry in cases.	Implemented both B1_Constant_Reversing (t \in [5.0, 30.0]\ \mathrm{s}, N_{\mathrm{eval}} = 751, 3 velocity reversals at t = 5.10, 15.10, 25.10\ \mathrm{s}) and B2_Constant_Unidirectional (t \in [15.5, 24.5]\ \mathrm{s}, N_{\mathrm{eval}} = 271, strictly \dot{q}_3 > 0, \dot{d}_4 > 0, 0 reversals) explicitly in m3_observer_suite.m and run_m3_experiments_v2.py. All report tables and figures are generated directly from the saved output (m3_audit_results.json

Audit ID	Audit Category	Root Cause in v1.0	Resolution in v2.0.0-PRELIMINARY
			/ M3_observer_audit_results.mat).
AUD-04	**Noise Provenance (Test Case G)**	Report v1.0 and code v1.0 referred to Test G as "G_Noisy (Sensor + Act. Noise)", implying sensor-calibrated noise levels when no sensor noise specifications exist in the repository.	Renamed and relabeled everywhere as **G_Synthetic_Noise_Stress (Uncalibrated Synthetic Gaussian Noise Stress Test)**, using a portable deterministic Park–Miller LCG + Box–Muller generator (seed = 202603) that produces identical realizations in MATLAB and Python.
AUD-05	**Branch B Uncertainty Bound Separation**	m3_observer_suite.m v1.0 printed only a single CovLag[%] column while Report v1.0 mixed raw and lag-compensated coverage in prose/tables.	Explicitly decomposed Branch B total error \tau_L - \hat{\tau}_{L,B} into (1) filtered friction discrepancy \hat{\tau}_{L,A} - \hat{\tau}_{L,B} (\vert{}\hat{F}_{c,\mathrm{filt},i}\vert{} \le f_{ci}), (2) FOH inter-sample chord error \tau_{L,\mathrm{filt}} - \hat{\tau}_{L,A}, and (3) dynamic observer tracking lag \tau_L - \tau_{L,\mathrm{filt}}. Both code and report now output three separate coverage columns: CovFricPct, CovLagPct, and CovRawPct.
AUD-06	**Task-Space Jacobian Factorization & Station Offset**	Report v1.0 §6 wrote J_P(q) = R(q_3)\,\mathrm{diag}(r_P, 1) and J_P(q)^{-T} =	Corrected factorization to J_s(q) = Q(q_3)\,\mathrm{diag}(s(d_4), 1) = R(q_3)\Pi\,\mathrm{dia

Audit ID	Audit Category	Root Cause in v1.0	Resolution in v2.0.0-PRELIMINARY
		R(q_3)\,\mathrm{diag}(1/r_P, 1), omitting the column-swap permutation matrix \Pi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} required because q = [q_3, d_4]^T orders the transverse revolute coordinate first and radial prismatic coordinate second. Also, m3_observer_suite.m v1.0 used r_bail = d4 + 2.05 m over d_4 \in [2.5, 4.5]\ \mathrm{m} instead of s_bail = d4 + 1.50 m over d_4 \in [6.75, 10.75]\ \mathrm{m} from recover_load.m line 17.	g}(s(d_4), 1) with \det(J_s) = -s(d_4), J_s(q)^{-T} = R(q_3)\Pi\,\mathrm{diag}(1/s(d_4), 1), and F_{\mathrm{equiv}} = \frac{\tau_3}{s}\hat{e}_\theta(q_3) + F_4\hat{e}_r(q_3) = R(q_3)[F_4;\ \tau_3/s]. Verified numerically against R.sig.Qq3_pred, R.sig.Qd4_pred, R.sig.Fx, and R.sig.Fy in M2_100kN_0p1111Hz_sp1p50.mat (s_{\mathrm{bail}} = d_4 + 1.50\ \mathrm{m} \in [8.2505, 12.2495]\ \mathrm{m}, inverse recovery error \le 2.91\times 10^{-11}\ \mathrm{N}).
AUD-07	**Reproducibility & PRELIMINARY Lock**	v1.0 lacked archive SHA-256 hashes, .slx provenance metadata, and portable RNG parity between MATLAB and Python.	Recorded full archive provenance (filename, SHA-256, R.prov.timestamp, R.prov.gitCommit, R.prov.matlab, N_raw=3001, fs_raw=100 Hz, TrimStart=5.0 s, k_frozen=20 s^-1, M=67, seed=202603, code_version=2.0.0-PRELIMINARY) and maintained **PRELIMINARY** status across all outputs pending frozen MS02P05 signals.

2. Quantitative Shift Log (v1.0 Synthetic vs.

v2.0.0-PRELIMINARY Actual Archive Ingestion)

Ingesting the actual M2_null.mat and M2_100kN_0p1111Hz_sp1p50.mat trajectories (d_4 \in [6.7505, 10.7495]\ \mathrm{m}, q_3 \in [30^\circ, 60^\circ], t \in [5.0, 30.0]\ \mathrm{s}, N_{\mathrm{eval}} = 751 at 30\ \mathrm{Hz}) produced the following verified shifts relative to v1.0:

- **Test Case A (A_Zero_M2_null, direct M2_null.mat)**:
  - Saddle Torque (\tau_{L,3}) RMSE shifted from 55.59\ \mathrm{N\cdot m} (v1.0) to 53.47\ \mathrm{N\cdot m} (Branch A) and from 297.93\ \mathrm{N\cdot m} to 300.43\ \mathrm{N\cdot m} (Branch B).
  - Crowd Force (F_{L,4}) RMSE shifted from 4.24\ \mathrm{N} (v1.0) to 4.36\ \mathrm{N} (Branch A) and from 473.52\ \mathrm{N} to 472.84\ \mathrm{N} (Branch B).
  - In M2_null.mat, the velocity zero-crossings occur at t = 5.10, 15.10, 25.10\ \mathrm{s} (which coincide with exact 30\ \mathrm{Hz} sample nodes k = 153, 453, 753 where \vert{}\dot{q}_3\vert{} = 2.12\times 10^{-7}\ \mathrm{rad/s} \ll \epsilon_3 and \vert{}\dot{d}_4\vert{} = 1.62\times 10^{-6}\ \mathrm{m/s} \ll \epsilon_4), reducing the zero-load during-reversal peak in Branch A from 38.85\ \mathrm{N\cdot m} (caused by the artificial phase shift in v1.0) to 7.10\ \mathrm{N\cdot m} (q_3) and 0.26\ \mathrm{N} (d_4).
- **Test Cases B1 & B2 (B1_Constant_Reversing vs. B2_Constant_Unidirectional on M2_null.mat)**:
  - Over the full reversing window t \in [5.0, 30.0]\ \mathrm{s} (B1), sign reversals partially cancel Branch B's unmodeled friction, giving a mean bias of +51.23\ \mathrm{N\cdot m} (q_3) and +89.36\ \mathrm{N} (d_4) alongside an MAE of 297.62\ \mathrm{N\cdot m} and 471.62\ \mathrm{N}.
  - Over the strictly unidirectional window t \in [15.5, 24.5]\ \mathrm{s} (B2, N_{\mathrm{eval}} = 271, \dot{q}_3 > 0, \dot{d}_4 > 0), Branch B's mean bias is exposed as -304.10\ \mathrm{N\cdot m} (q_3, comprising -340.00\ \mathrm{N\cdot m} Coulomb friction +35.90\ \mathrm{N\cdot m} dynamic FOH chord offset) and -475.69\ \mathrm{N} (d_4, comprising -480.00\ \mathrm{N} Coulomb friction +4.31\ \mathrm{N} dynamic FOH chord offset), compared to +35.90\ \mathrm{N\cdot m} and +4.31\ \mathrm{N} in Branch A.
- **Test Case C (C_Sinusoidal_M2_100kN, direct M2_100kN_0p1111Hz_sp1p50.mat)**:
  - Because M2_100kN_0p1111Hz_sp1p50.mat applies F_y(t) = 100\ \mathrm{kN}\sin(2\pi t/9) at the bail station s(t) = d_4(t) + 1.50\ \mathrm{m} \in [8.2505, 12.2495]\ \mathrm{m}, the true generalized loads Q_{q_3,\mathrm{pred}}(t) = s(t)F_y(t)\cos q_3(t) and Q_{d_4,\mathrm{pred}}(t) = F_y(t)\sin q_3(t) have peaks of 734.24\ \mathrm{kN\cdot m} and 86.52\ \mathrm{kN} (vs. the synthetic 245\ \mathrm{kN\cdot m} and 49\ \mathrm{kN} in v1.0).
  - Consequently, at k_3 = k_4 = 20\ \mathrm{s}^{-1}, Branch A's total tracking RMSE is 16{,}694.23\ \mathrm{N\cdot m} (2.27\% of peak, phase lag 50.05\ \mathrm{ms} = 2.00^\circ, amplitude error -0.065\%) on q_3 and 1{,}766.46\ \mathrm{N} (2.04\% of peak, phase lag 50.05\ \mathrm{ms}, amplitude error -0.070\%) on d_4.
- **Test Case G (G_Synthetic_Noise_Stress on M2_100kN_0p1111Hz_sp1p50.mat)**:
  - Because d_4 \in [6.75, 10.75]\ \mathrm{m} in M2_100kN_0p1111Hz_sp1p50.mat yields D_{11}(d_4) = I_{zz} + M_d(d_4 - 1.32)^2 \in [1.89\times 10^6,\ 5.12\times

10^6]\ \mathrm{kg\cdot m^2} (vs. \sim 5.5\times 10^5\ \mathrm{kg\cdot m^2} when d_4 \in [2.6, 4.1]\ \mathrm{m}), the uncalibrated synthetic velocity perturbation \sigma_{\dot{q}_3} = 3\ \mathrm{mrad/s} amplified by k_3 D_{11}(d_4) produces a saddle RMSE of 180.05\ \mathrm{kN\cdot m} at k = 20\ \mathrm{s}^{-1} (and 4.74\ \mathrm{kN} on d_4 where D_{22} = M_d = 54{,}300\ \mathrm{kg} is constant).

- **Task-Space Jacobian (run_jacobian_analysis on M2_100kN_0p1111Hz_sp1p50.mat)**:
  - Across the actual archive stroke d_4 \in [6.7505, 10.7495]\ \mathrm{m}, the COG lever arm is s_{\mathrm{COG}} = d_4 - 1.32 \in [5.4305, 9.4295]\ \mathrm{m} (\kappa_2(J_{\mathrm{COG}}) \in [5.43, 9.43]) and the bail lever arm is s_{\mathrm{bail}} = d_4 + 1.50 \in [8.2505, 12.2495]\ \mathrm{m} (\kappa_2(J_{\mathrm{bail}}) \in [8.25, 12.25]), yielding an equivalent bail tip-force Coulomb friction floor \Vert{}J_{\mathrm{bail}}^{-T} F_c\Vert{}_2 = \sqrt{(340/s_{\mathrm{bail}})^2 + 480^2} \in [480.80,\ 481.77]\ \mathrm{N}.